$V = \text{Joule}/\text{coloumb}$

$I = \text{coloumb}/\text{sec}$

$P = VI = I^2R = \frac{V^2}{R} = \frac{J}{C} \times \frac{C}{s} = \text{Joule}/\text{sec} = \text{watt}$

Ohm's law:

Ohm's law states that the voltage across a resistor is directly proportional to the current flowing through the resistor.

$V \propto I$

$\therefore V = IR$

$I \propto V$

or, $I = GV$

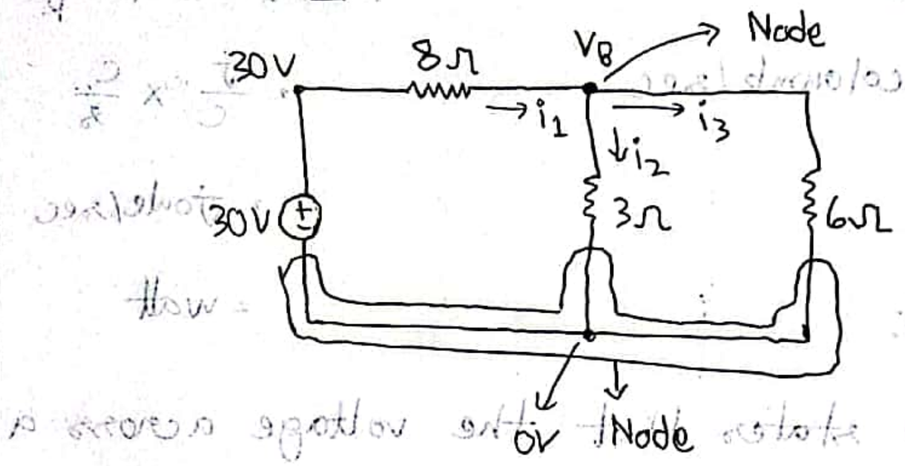
$\therefore I = \frac{V}{R}$ $[G = \frac{1}{R}]$

$\Omega = \frac{1}{S} \rightarrow$ সিমেন্স

$G \rightarrow$ Conductance

$P \rightarrow$ Power

KCL (Kirchhoff's Current Law):



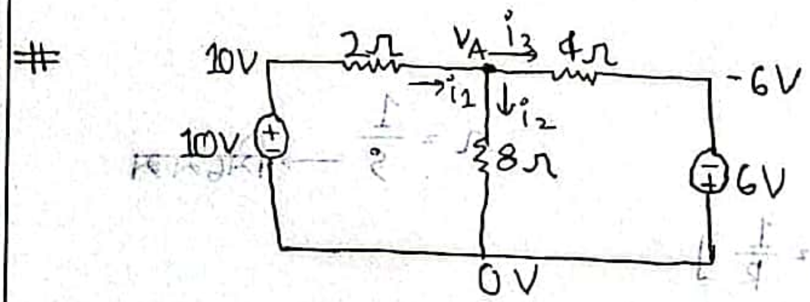
$i_1 = i_2 + i_3$

or, $\frac{30 - V_B}{8} = \frac{V_B - 0}{3} + \frac{V_B - 0}{6}$

or, $V_B \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{8}\right) = \frac{30}{8}$

$\therefore V_B = 6$

#



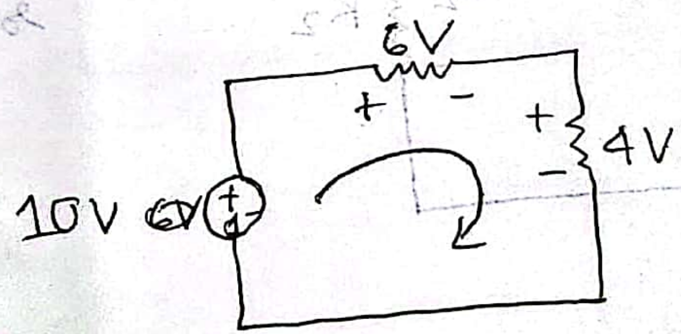
$i_1 = \frac{10 - V_A}{2}$ $\quad i_3 = \frac{V_A + 6}{4}$ $\quad i_2 = \frac{V_A - 0}{8}$

$i_1 = i_2 + i_3$

$\therefore V_A = 4$

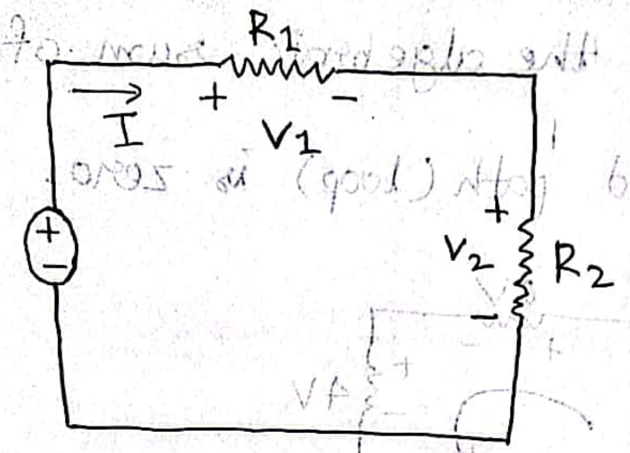
KVL (Kirchhoff's Voltage Law):

KVL states that the algebraic sum of all voltages around a closed path (loop) is zero.



$$+6V + 4V - 10V = 0$$

Voltage Divider Rules (VDR):



Series 9 current same.

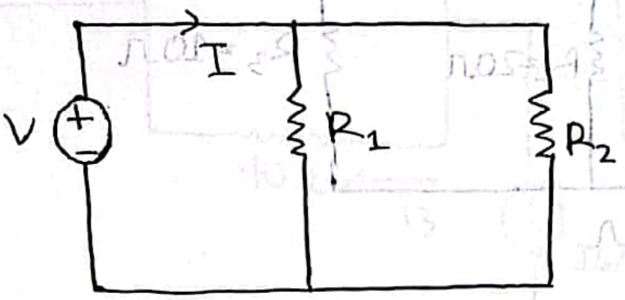
$$I = \frac{V}{R_1+R_2}$$

$$V_1 = IR_1$$

$$\text{or, } V_1 = \frac{V}{R_1+R_2} R_1$$

$$\therefore V_1 = \left(\frac{R_1}{R_1+R_2}\right)V$$

Current Divider Rules (CDR):



Parallel 𝑔 Voltage same.

$$I = \frac{V}{R_{eq}}$$

$$\left[\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}\right]$$

$$\therefore R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{V}{R_1}$$

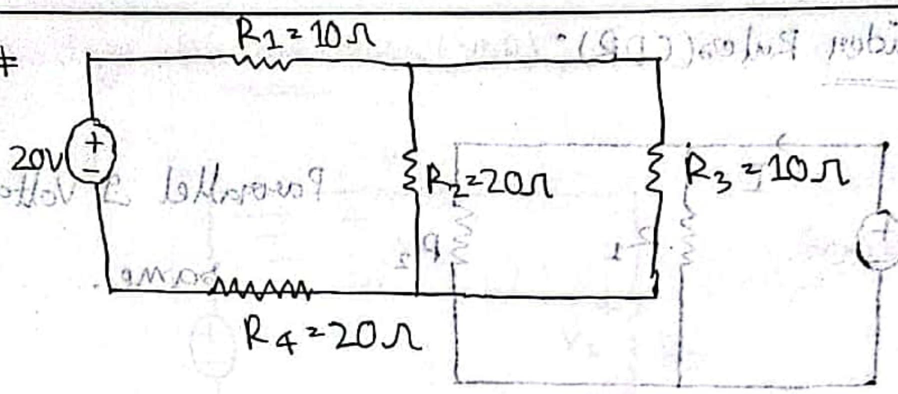
$$= \frac{I R_{eq}}{R_1}$$

$$= \frac{I}{R_1} \cdot \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore I_1 = I\left(\frac{R_2}{R_1 + R_2}\right)$$

$$I_2 = I\left(\frac{R_1}{R_1 + R_2}\right)$$

#



$R_T = 36.67\,\Omega$

$V = 20\,V$

$I = \frac{V}{R}$

$= 0.5454A = I_1 = I_4$

$I_2 = \frac{R_3}{R_3 + R_2} I$

$= 0.1818\,A$

$I_3 = \frac{R_2}{R_2 + R_3} I$

$= 0.3636A$

$V_1 = IR_1$

$= 5.454V$

$V_2 = I_2 R_2$

$= 3.636V$

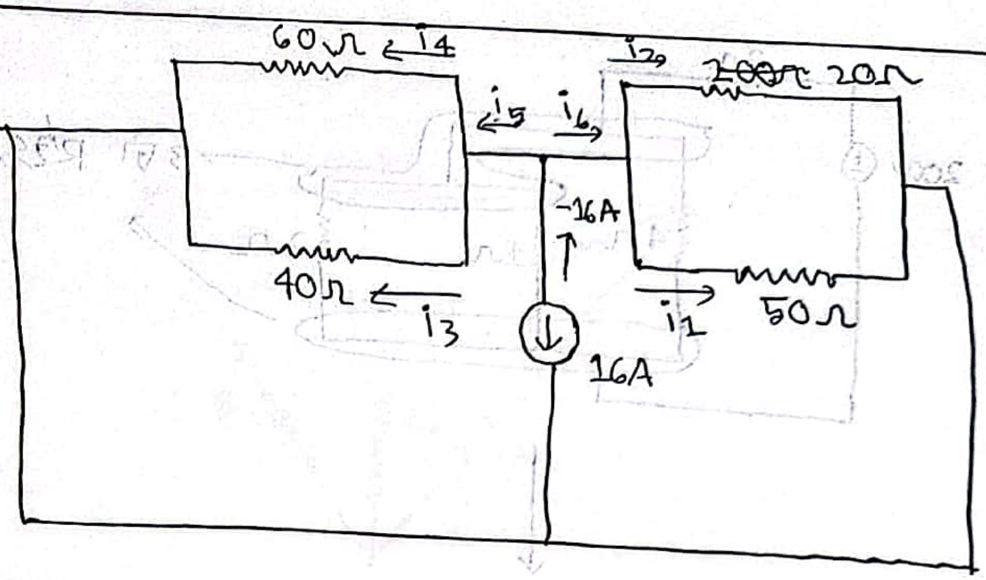
$= V_3$

$V_4 = IR_4$

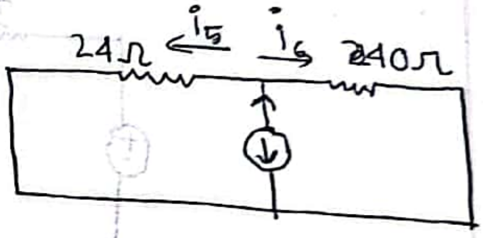
$= 0.5454 \times 20$

$= 10.908V$

#



$I = -16A$



$$I_6 = \frac{24}{24+40}(-16)$$

$$= -6$$

$$I_5 = -10$$

$$I_4 = \frac{40}{40+60}(-10)$$

$$= 4$$

$$I_3 = -6$$

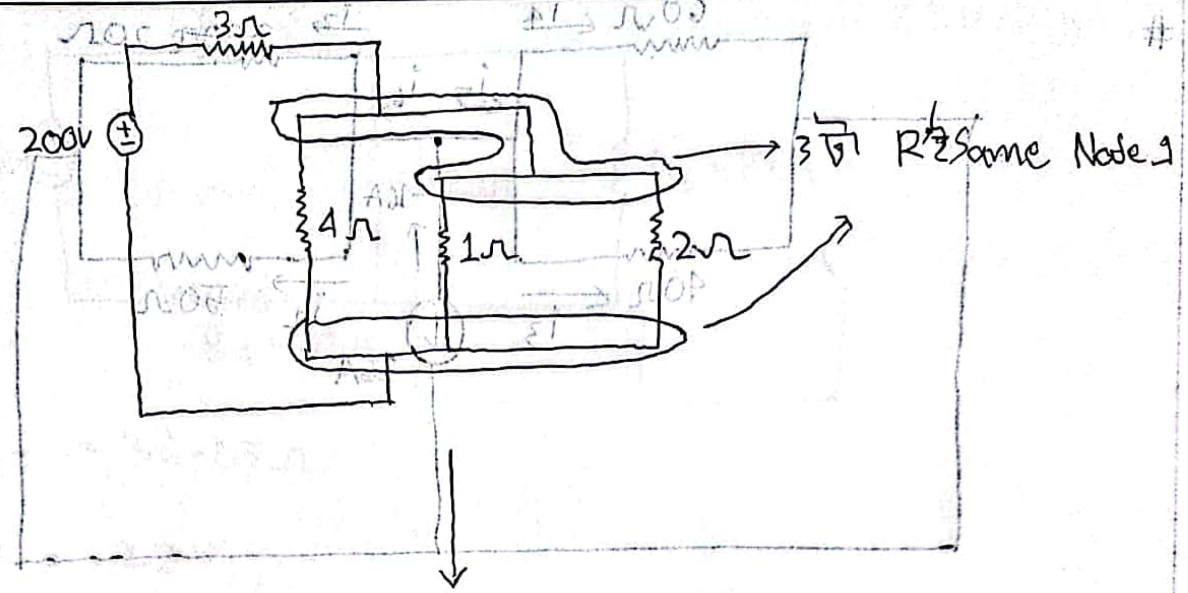
$$I_2 = \frac{50}{50+20}(-6)$$

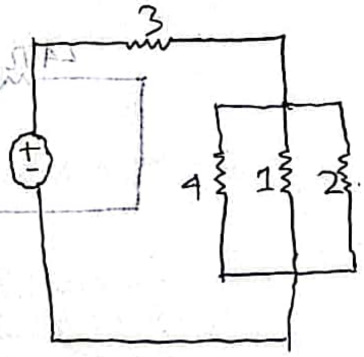
$$I_1 = \frac{20}{50+20}(-6)$$

V ⊕ → Independent Voltage Source

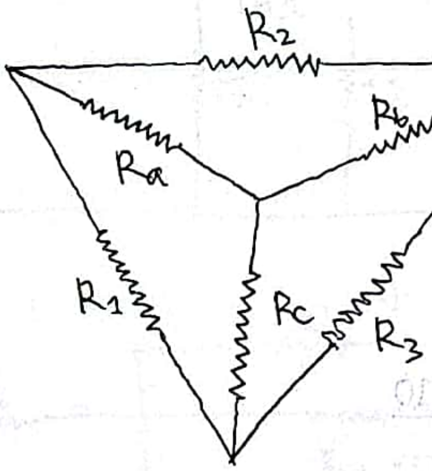
I ⓘ → Inde Current Source

#


200V
3Ω
4Ω
1Ω
2Ω
3 টা R? Same Node 1


3
4
1
2

Y ⟶ Δ:



$$R_1 = \frac{R_aR_b + R_bR_c + R_cR_a}{R_b}$$

$$R_2 = \frac{R_aR_b + R_bR_c + R_cR_a}{R_c}$$

$$R_3 = \frac{R_aR_b + R_bR_c + R_cR_a}{R_a}$$

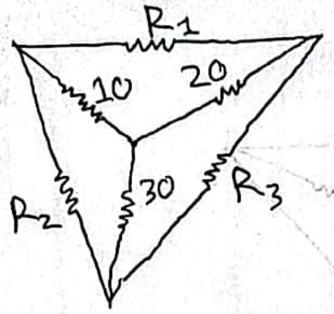
Δ ⟶ Y:

$$R_a = \frac{R_1R_2}{R_1 + R_2 + R_3}$$

$$R_b = \frac{R_2R_3}{R_1 + R_2 + R_3}$$

$$R_c = \frac{R_1R_3}{R_1 + R_2 + R_3}$$

#



Ans:

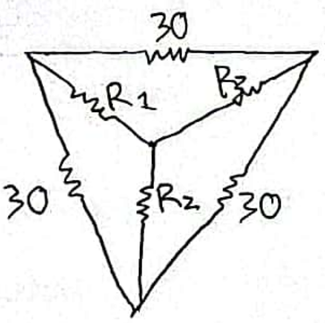
$$R_1 = \frac{10\cdot 20+20\cdot 30+30\cdot 10}{30}$$

$$= \frac{1100}{30}$$

$$R_2 = \frac{1100}{20}$$

$$R_3 = \frac{1100}{10}$$

#

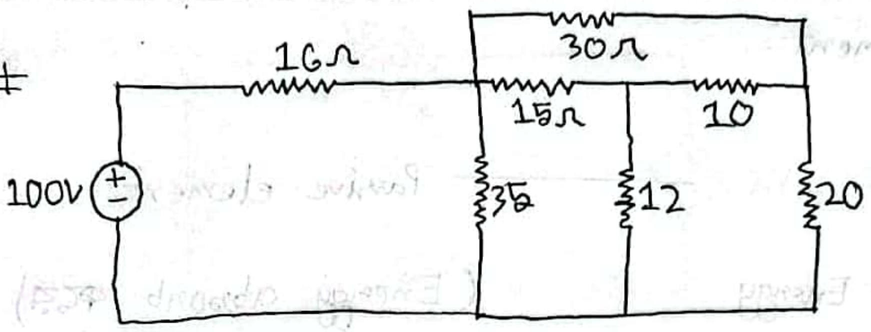


$$R_1 = \frac{30\times 30}{30+30+30} = \frac{900}{90}$$

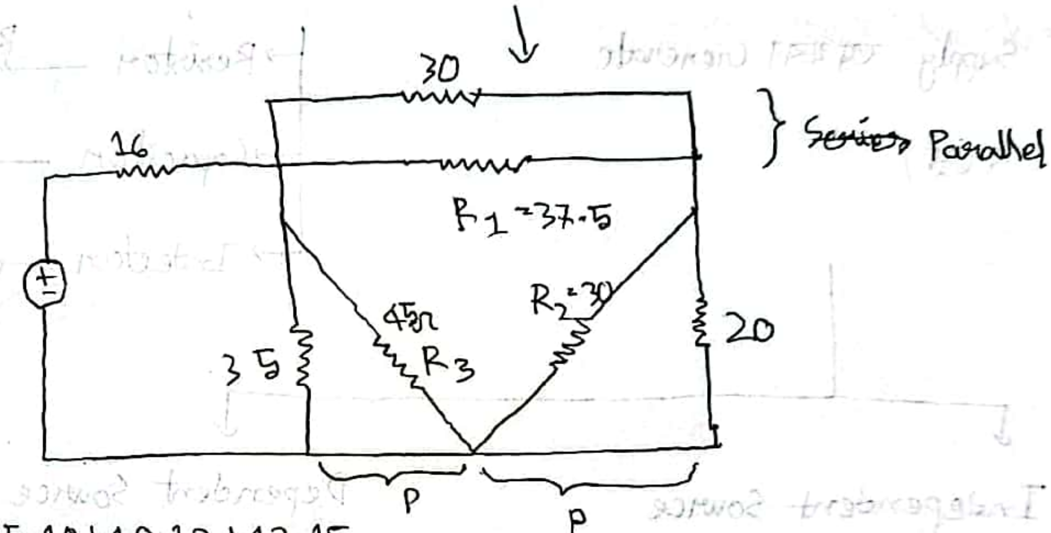
$$R_2 = \frac{900}{90}$$

$$R_3 = \frac{900}{90}$$

#



Ans:



$$R_1 = \frac{15\cdot 10 + 10\cdot 12 + 12\cdot 15}{12} = 37.5$$

$$R_2 = \frac{450}{15} = 30$$

$$R_3 = \frac{450}{10} = 45$$

$R_T = 16 +$

$$\frac{1}{R_{P_1}} = \frac{1}{30} + \frac{1}{37.5}$$

$$\therefore R_{P_1} = 16.67$$

$$\frac{1}{R_{P_2}} = \frac{1}{R_2} + \frac{1}{20}$$

$$\therefore R_{P_2} = 12$$

$$\frac{1}{R_{P_3}} = \frac{1}{R_3} + \frac{1}{35}$$

$$\therefore R_{P_3} = 19.6875$$

$$R_S = R_{P_2} + R_{P_3} = 31.7$$

$$\frac{1}{R_P} = \frac{1}{R_{P_1}} + \frac{1}{R_S}$$

Electric Element:

Active element

(যেসকল Element Energy Supply অথবা Generate করে)

Passive element

(Energy absorb করে)

- Resistor — R
- Capacitor — C
- Inductor — L

Independent Source

- Voltage source
- Current source

Dependent Source

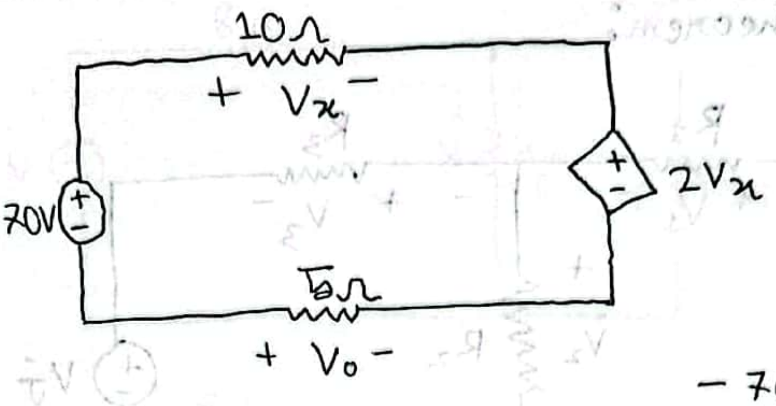
Voltage Source

- Voltage dependent on Voltage source — $2V_o$
- Current dependent on Voltage source — $2i_o$

Current Source

- Voltage dependent on current source — xV_o
- Current dependent on current source — xi_o

#



$V_x = IR = i \cdot 10$

$2V_x = 20i$

$-70V + V_x + 2V_x - V_o = 0$

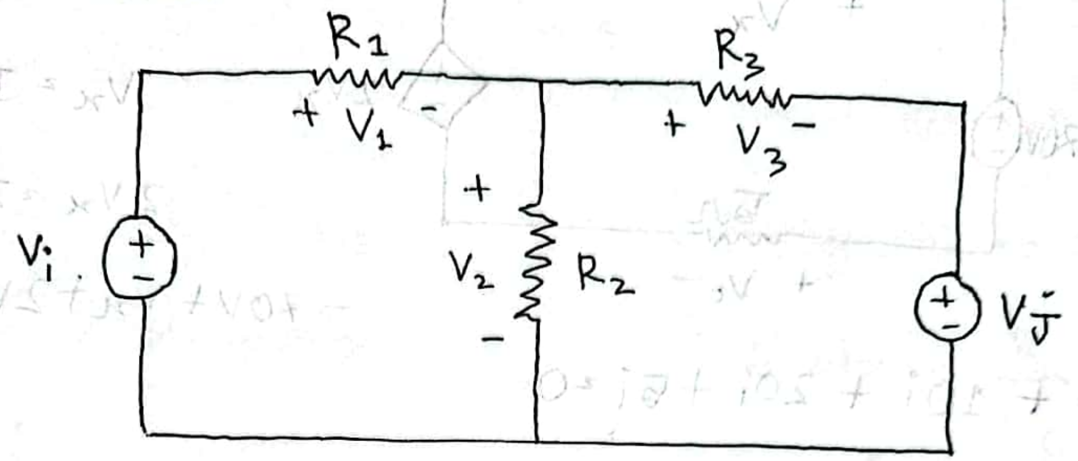
$-70 + 10i + 20i + 5i = 0$

$\therefore i = 2A$

$\therefore V_x = 20$

$\therefore V_o = -10 \quad [-V_o = 5i \therefore V_o = -5i]$

Superposition Theorem:



$V_2 = V_2' + V_2''$

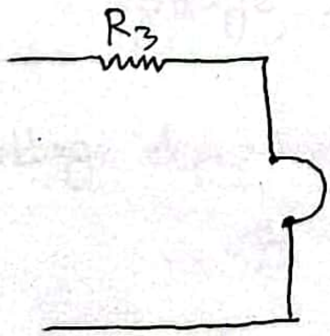
$V_2'' \rightarrow V_i$ বাদ দিয়ে

$V_2' \downarrow$ V_j বাদ দিয়ে হিসাব

Inactive

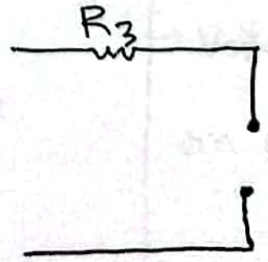
- → Voltage Source (Short করা / তার দিয়ে যুক্ত করা)
- → Current " (Open)

V.S:

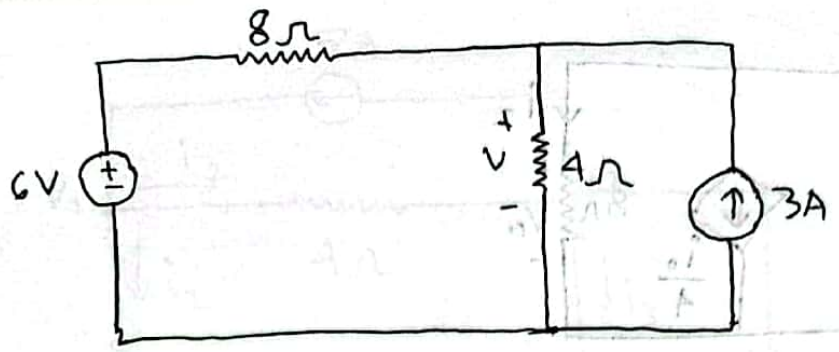


→ Connecting Wire

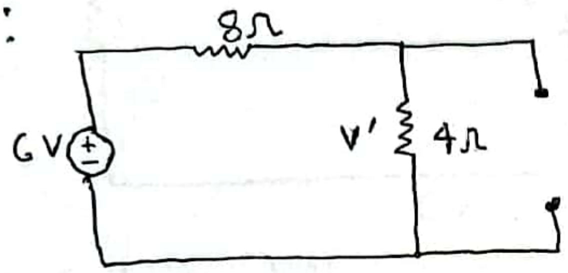
C.S:

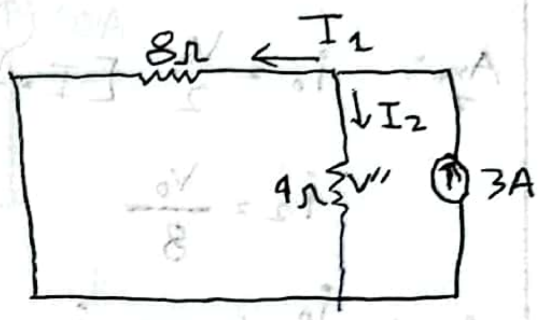


#



Ans:





With VDR and CDR:

$V' = \frac{4}{4+8} \times 6$

$= 2V$

$V'' = I_2 R_2$

$= 2 \times 4$

$= 8V$

$\therefore V = V' + V''$

$= 10V$

$I_1 = \frac{4}{8+4} \times 3$

$= 1A$

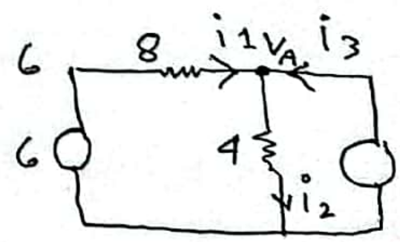
$I_2 = 2A$

———o———

With KCL:

$i_1 = \frac{6 - V_A}{8}$

$i_2 = \frac{V_A - 0}{4}$

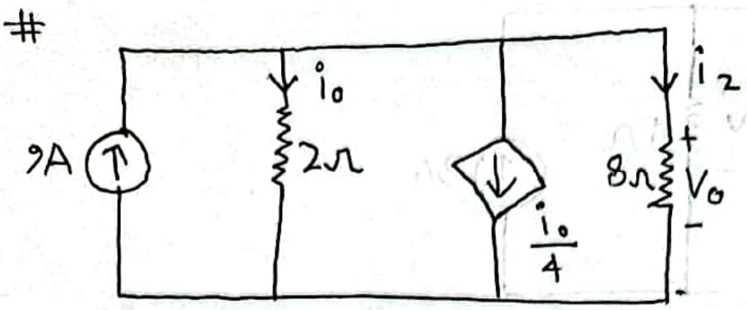


$i_1 + i_3 - i_2 = 0$

or, $\frac{6}{8} - \frac{V_A}{8} + 3 - \frac{V_A}{4} = 0$

$\therefore V_A = 10V$

#



Ans: $i_0 = \frac{V_0}{2}$ $[I = \frac{V}{R}]$

$i_2 = \frac{V_0}{8}$

$i_0 + \frac{i_0}{4} + i_2 = 9$

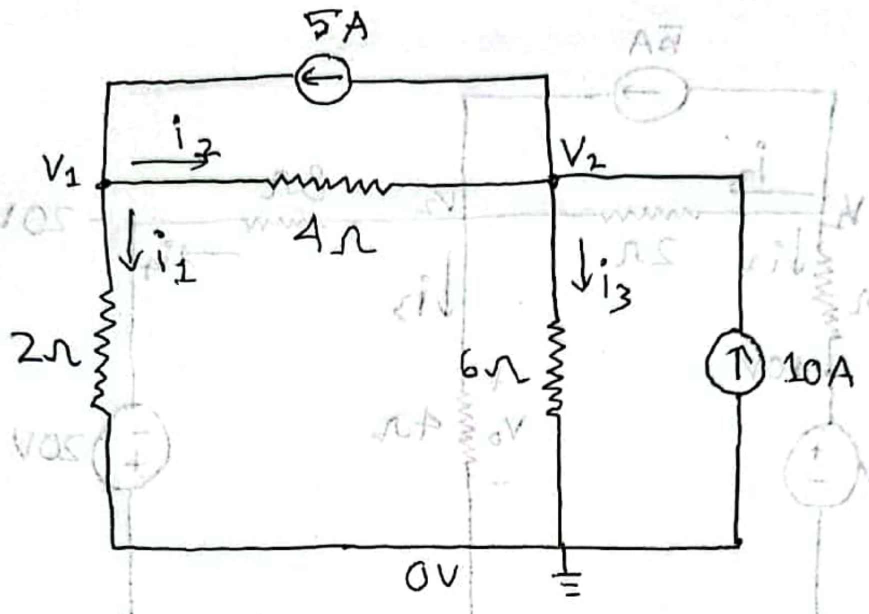
or, $i_0 + i_0/4 + \frac{V_0}{8} = 9$

or, $i_0 + \frac{i_0}{4} + \frac{2i_0}{8} = 9$ $[i_0 = \frac{V_0}{2} \therefore V_0 = 2i_0]$

or, $i_0(1 + \frac{1}{4} + \frac{1}{4}) = 9$

$\therefore i_0 = 6A$

#



Ans: Applying KCL,

$$i_1 = \frac{V_1 - 0}{2} \qquad i_2 = \frac{V_1 - V_2}{4} \qquad i_3 = \frac{V_2 - 0}{6}$$

$i_1 + i_2 = 5$

or, $\frac{V_1}{2} + \frac{V_1}{4} - \frac{V_2}{4} = 5$

$\therefore V_1\left(\frac{1}{2} + \frac{1}{4}\right) - V_2 \cdot \frac{1}{4} = 5$

$i_2 - i_3 + 10 - 5 = 0$

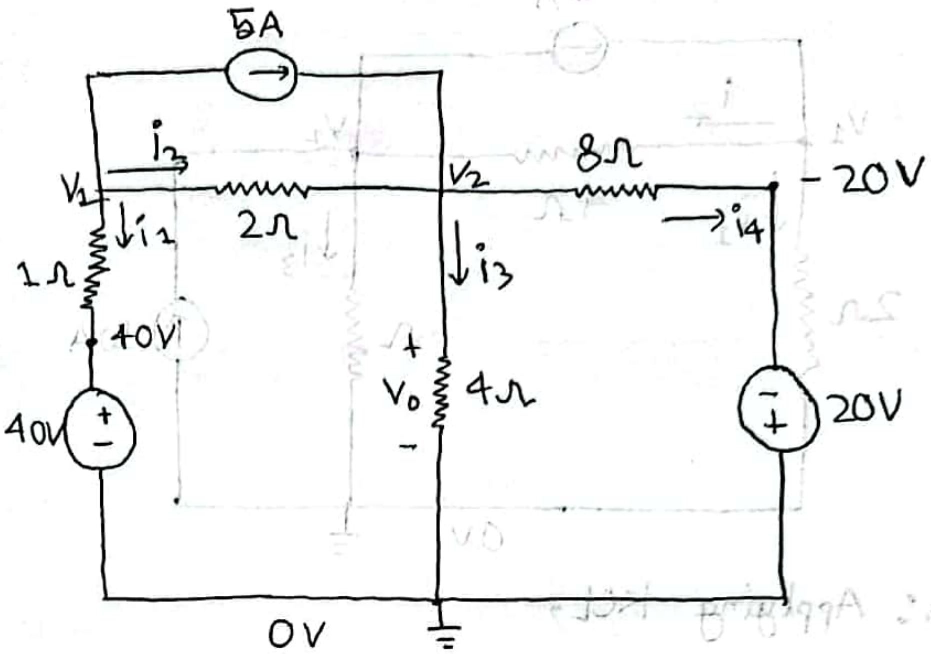
or, $\frac{V_1}{4} - \frac{V_2}{4} - \frac{V_2}{6} = -5$

or, $V_1 \frac{1}{4} - V_2\left(\frac{1}{4} + \frac{1}{6}\right) = -5$

$\therefore V_1 = 13.333V$

$\therefore V_2 = 20V$

(3.14)



$i_1 = \frac{V_1-40}{1}$ $\quad i_2 = \frac{V_1-V_2}{2}$ $\quad i_3 = \frac{V_2-0}{4}$ $\quad i_4 = \frac{V_2+20}{8}$

$i_1+i_2+5=0$ $\qquad i_2+5-i_3-i_4=0$

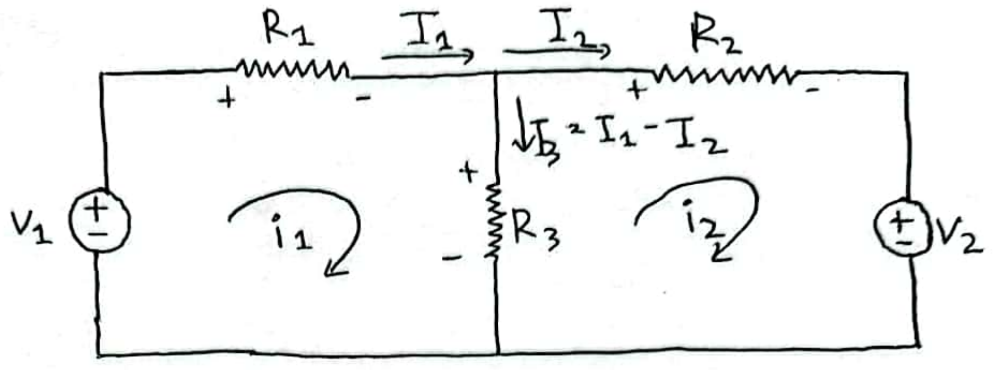
or, $\frac{V_1}{1} - 40 + \frac{V_1}{2} - \frac{V_2}{2} = -5$ $\qquad$ or, $\frac{V_1}{2} - \frac{V_2}{2} + 5 - \frac{V_2}{4} - \frac{V_2}{8} - \frac{20}{8} = 0$

$\therefore V_1\left(1+\frac{1}{2}\right) - V_2\cdot\frac{1}{2} = 35$ $\qquad \therefore V_1\cdot\frac{1}{2} - V_2\cdot\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right) = \frac{20}{8} - 5$

$\therefore V_1 = 30V$

$\therefore V_2 = 20V$

#



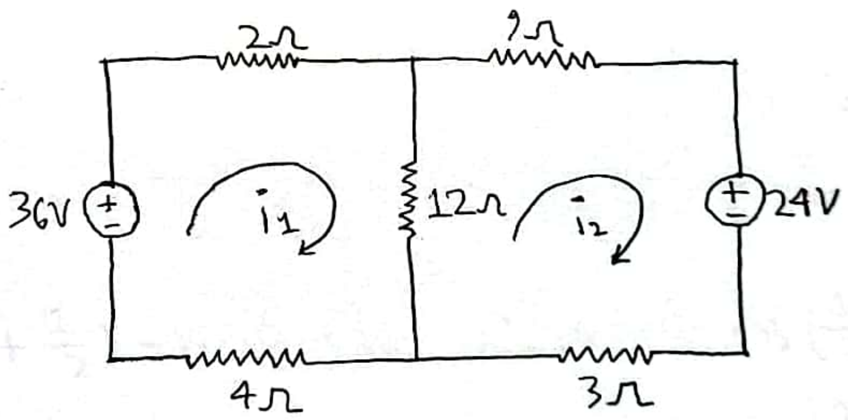
Ans: $-V_1 + I_1R_1 + I_1R_3 - I_2R_3 = 0$

$\therefore I_1R_1 + I_1R_3 - I_2R_3 = -V_2$

Loop-2:

$\therefore -I_1R_3 + I_2R_2 + I_2R_3 = -V_2$

(3.5)



$i_1(2+12+4) - i_2(12) = 36$ [Loop i_1]

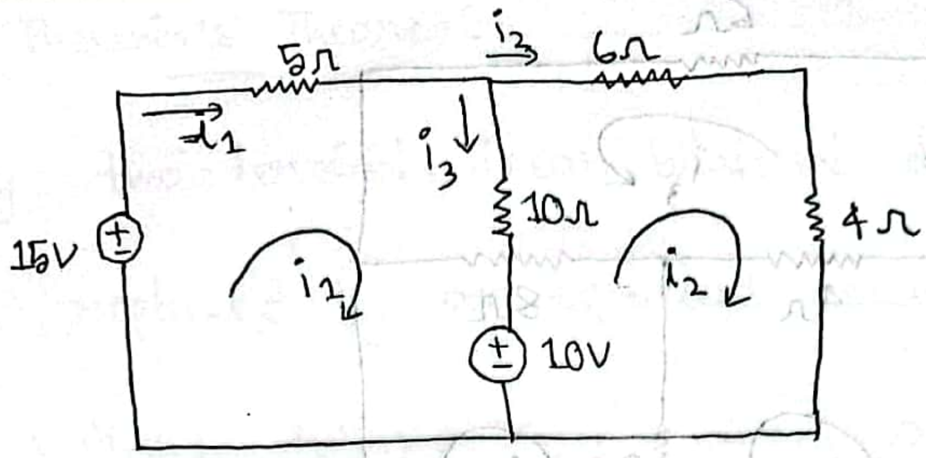
$-i_1(12) + i_2(3+12+9) = -24$ [Loop i_2]

— o —

i_1(মোট রোধ) − i_2(কমন রোধ) = ভোল্টেজ — (i)

−i_1(কমন রোধ) + i_2(মোট রোধ) = ভোল্টেজ — (ii)

#



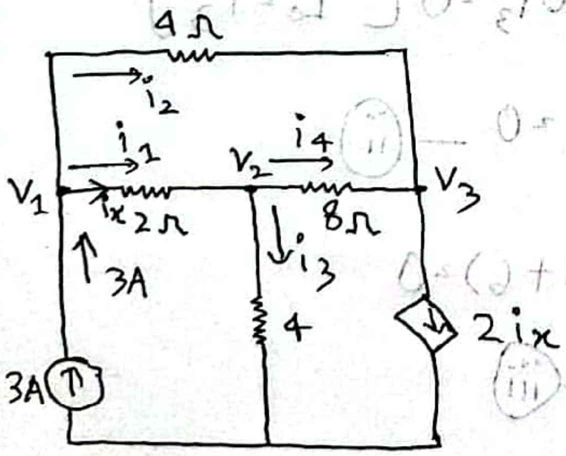
$$i_1(5+10) - i_2(10) = 15 - 10$$

$$-i_1(10) + i_2(10+6+4) = 10$$

$\therefore i_1 = 1A$

$\therefore i_2 = 1A$

Ex(3.2)



$$i_1 = i_x = \frac{V_1 - V_2}{2}$$

$$i_2 = \frac{V_1 - V_3}{4}$$

$$i_3 = \frac{V_2}{4}$$

$$i_4 = \frac{V_2 - V_3}{8}$$

$$i_1 + i_2 = 3$$

$$\therefore V_1\left(\frac{1}{2} + \frac{1}{4}\right) - V_2 \cdot \frac{1}{2} - V_3 \cdot \frac{1}{4} = 3$$

$$i_1 - i_3 - i_4 = 0$$

$$\therefore V_1 \cdot \frac{1}{2} - V_2\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) + V_3 \cdot \frac{1}{8} = 0$$

$$i_4 + i_2 - 2i_1 = 0 \; [2i_x = 2i_1]$$

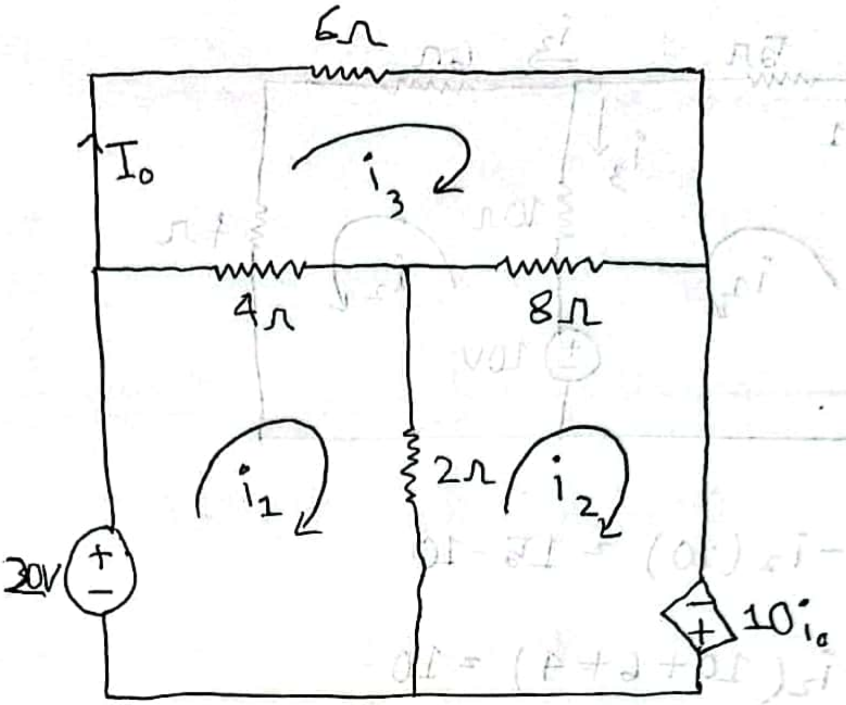
$$\therefore V_1\left(\frac{1}{4} - 1\right) + V_2\left(\frac{1}{8} + 1\right) + V_3\left(-\frac{1}{8} - \frac{1}{4}\right) = 0$$

$V_1 = 4.8V$

$V_2 = 2.4V$

$V_3 = -2.4V$

(3.6)



$$i_1(2+4) - i_2(2) - i_3(4) = \cancel{30}\ 20$$

$$\therefore 6i_1 - 2i_2 - 4i_3 = 20 \quad \text{— (i)}$$

$$-i_1(4) + i_2(2+8) - i_3(8) - 10i_3 = 0 \quad [I_0 = i_3]$$

$$\therefore -4i_1 + 10i_2 - 8i_3 - 10i_3 = 0 \quad \text{— (ii)}$$

$$-i_1(4) \cancel{+} - i_2(8) + i_3(4+8+6) = 0$$

$$\therefore -4i_1 - 8i_2 + 18i_3 = 0 \quad \text{— (iii)}$$

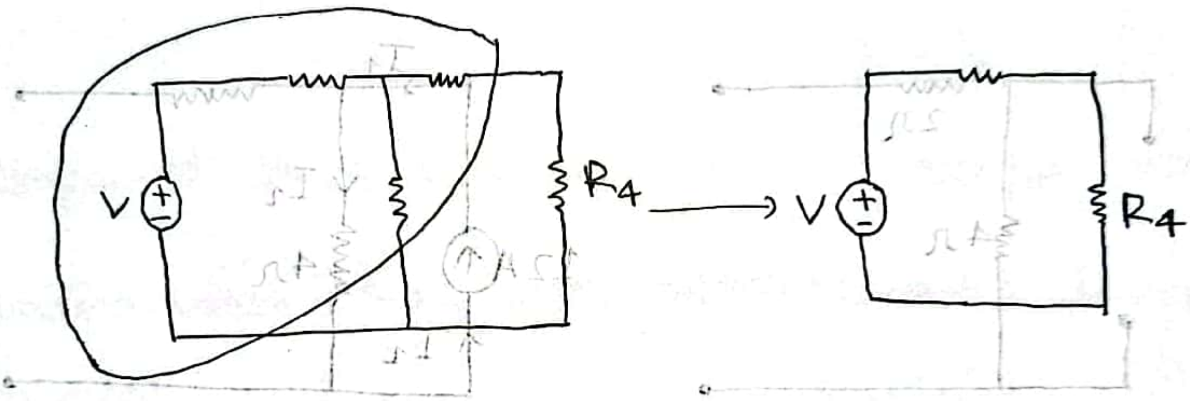
$i_1 = -3.21A$

$i_2 = -9.643$

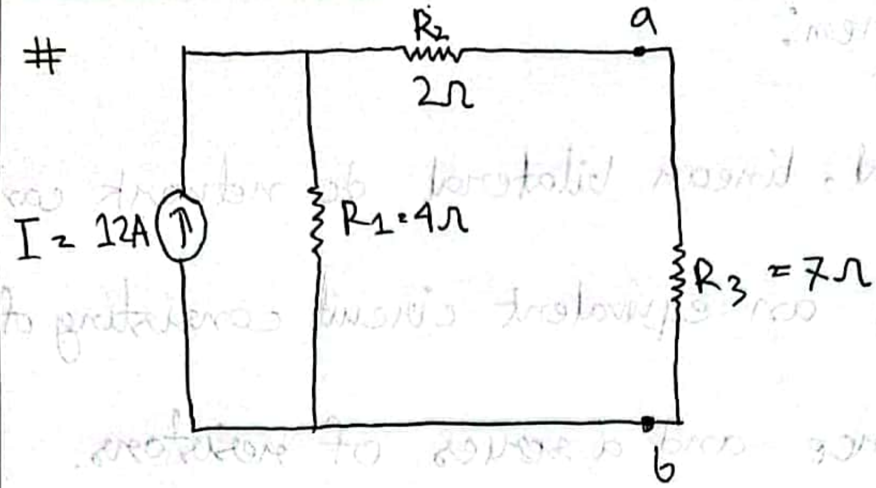
$i_3 = -5$

Thevenin's Theorem:

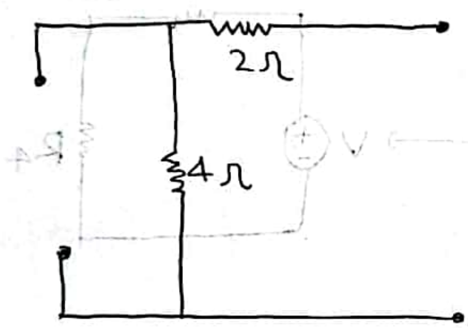
Any two-terminal, linear bilateral dc network can be replaced by an equivalent circuit consisting of a voltage source and a series of resistors.

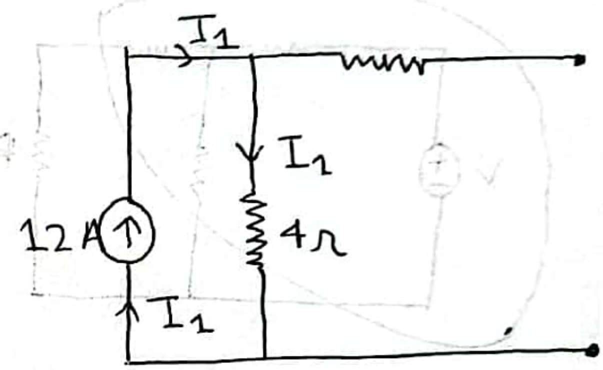


#



Ans:



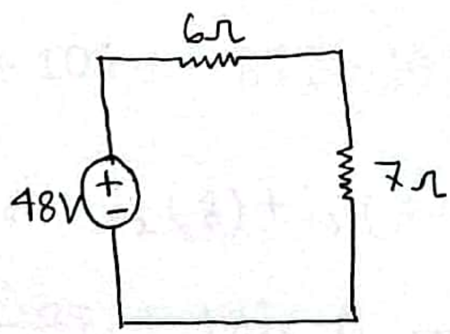


$R_{TH} = 2+4 = 6Ω$

$V_{TH} = I_1 R_1$

$= 12 \cdot 4$

$= 48V$



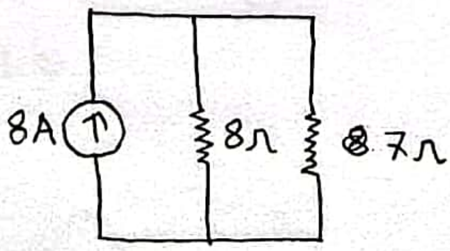
→ Thevenin's theorem

Ans.

Source transformation -

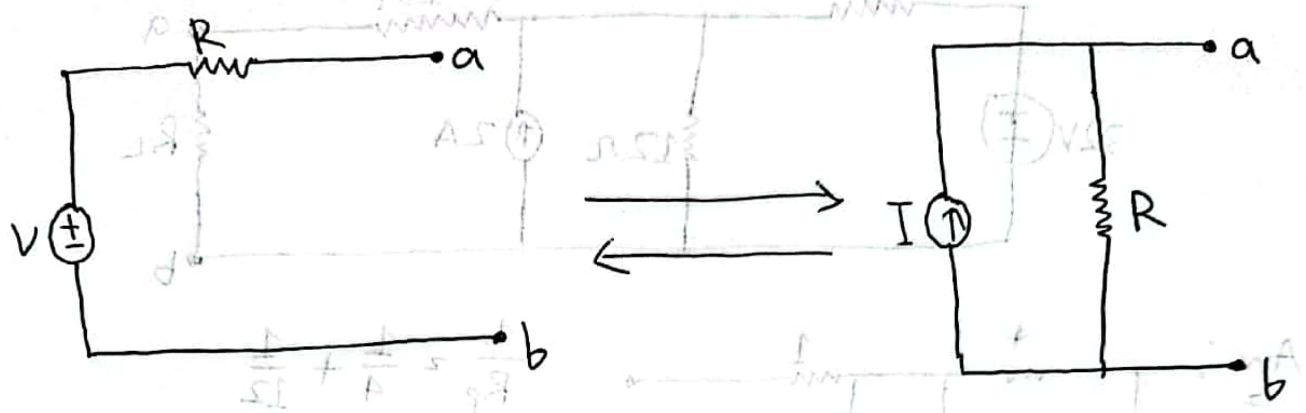
$I = \frac{V}{R_{TH}}$

$= 8A$



→ Norton's theorem

Source Transformation:

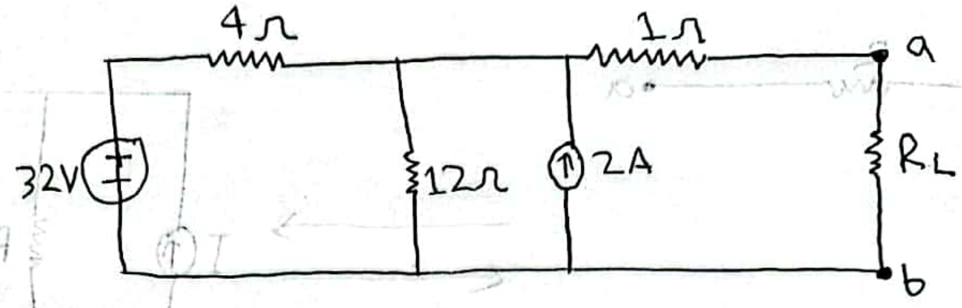


$$I = \frac{V}{R}$$

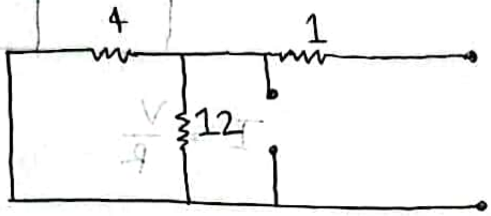
Voltage এর সাথে Resistor series এ থাকলে তাকে source transformation করা যাবে, করলে Current source এর resistor parallel এ হবে,

[V এর সাথে R Parallel এ হবে না]

Ex-4.8



Ans:

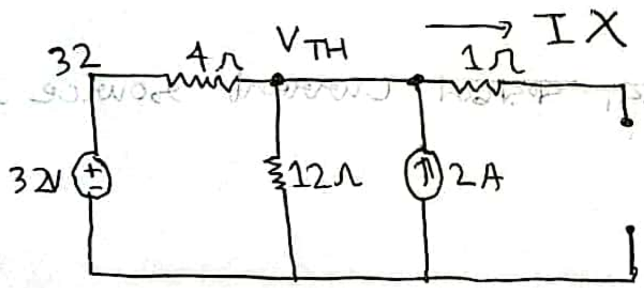


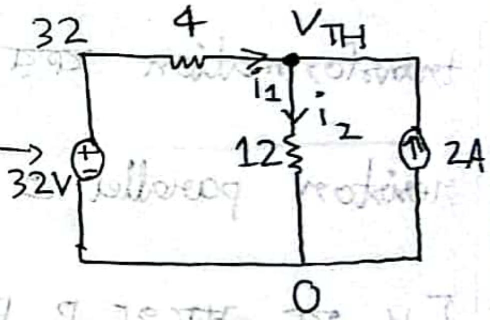
$$\frac{1}{R_p} = \frac{1}{4} + \frac{1}{12}$$

$$\therefore R_p = 3$$

$$\therefore R_{TH} = 4\Omega$$

Ans.





$$i_1 + 2 = i_2$$

$$\text{or, } \frac{32 - V_{TH}}{4} + 2 = \frac{V_{TH} - 0}{12}$$

$$\therefore V_{TH} = 30V$$

Ans.